

Andrew P. Grace

ASSISTANT PROFESSOR OF APPLIED MATHEMATICS

Department of Applied Mathematics · University of Colorado Boulder, Boulder, CO, 80309, U.S.A.

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Professional Appointments

Assistant Professor of Applied Mathematics

UNIVERSITY OF COLORADO, BOULDER

Boulder, Colorado, U.S.A.

August 2025 - Present

NSERC Postdoctoral Research Fellow

HELD AT THE UNIVERSITY OF NOTRE DAME

Notre Dame, Indiana, U.S.A.

January 2025 - August 2025

ND-ECI Postdoctoral Research Fellow

HELD AT THE UNIVERSITY OF NOTRE DAME

Notre Dame, Indiana, U.S.A.

September 2023 - August 2025

Education

University of Waterloo

PHD IN APPLIED MATHEMATICS (WATER)

Waterloo, Ontario, Canada

September 2018 - August 2022

- Advisors: Marek Stastna, Kevin Lamb, and Andrea Scott
- **2022 Winner of the Tertia M.C. Hughes Prize** (Canadian Meteorological and Oceanographic Society)

University of Waterloo

M.MATH IN APPLIED MATHEMATICS

Waterloo, Ontario, Canada

September 2016 - August 2018

- Advisors: Marek Stastna, Francis Poulin

Trent University

B.SC MATHEMATICAL PHYSICS (HONS)

Peterborough, Ontario, Canada

September 2012 - May 2016

Publications

1. **Grace, A. P.**, & Richter, D. (2026). Incorporating particle settling enhancement in a coarse-mode deposition model. *Geophysical Research Letters*, 53, e2025GL119635.
2. Menicali, L., **Grace, A.**, Richter, D. H., & Castruccio, S. (2026). A Physics-Informed Spatiotemporal Deep Learning Framework for Turbulent Systems. *Journal of the American Statistical Association*, 1–20.
3. Swartz-Schult, K., **Grace, A.**, Bragg, A., Anderson, J. C., Cantrell, W., Grosz, R., & Richter, D. (2026). Non-Stokes deposition velocities observed under Rayleigh–Bénard turbulence. *Aerosol Science and Technology*, 1–18.
4. **Grace, A. P.**, Berk, T., Bragg, A. D., & Richter, D. H. (2025). Effects of settling on inertial particle slip velocity statistics in wall-bounded flows. *Journal of Fluid Mechanics*, 1015, A21.
5. **Grace, A. P.**, & Richter, D. (2025). Multi-scale interactions in turbulent mixed convection drive efficient transport of Lagrangian particles. *Journal of Fluid Mechanics*, 1008, A30.
6. **Grace, A. P.**, Richter, D. H., & Bragg, A. D. (2024). A Reinterpretation of Phenomenological Modeling Approaches for Lagrangian Particles Settling in a Turbulent Boundary Layer. *Boundary-Layer Meteorology*, 190(4), 1–31.
7. Castro-Folker, N., **Grace, A. P.**, and Stastna, M. (2023). Three-dimensional structure of cold-water gravity currents. *Physical Review Fluids*, 8(11), 113901.
8. **Grace, A. P.**, Fogal, A., and Stastna, M. Restratification in late winter lakes induced by cabbeling. *Geophysical Research Letters*, 50(14), e2023GL103402. (2023) (**Featured on the Water Institute homepage**)

9. **Grace, A.P.**, Stastna, M., Lamb, K.G. and Scott, K.A. Gravity currents in the cabbelling regime. *Physical Review Fluids* 8(1) (2023):014502 (**Featured in Physics Magazine** and **Editor's Highlight**)
10. Legare, S., **Grace, A.P.**, and Stastna, M. Double Diffusive Instability with a Constriction. *Physics of Fluids* 35(2) 024109 (2023) (**Scilight**)
11. Stastna, M., **Grace, A.P.**, and Robinson, T. Exact solutions for flow through porous media with the Klinkenberg Effect. *AIP Advances* 13, (2023):015309
12. Allum, D.J.M., **Grace, A.P.**, and Stastna, M. Two dimensional simulations of flow in ice-covered lakes with horizontal variations in surface albedo. *Physical Review Fluids* 7, (2022):103501
13. Stastna, M., Deepwell, D. and **Grace, A.** Shear instability in mode-2 internal Kelvin waves. *Environmental Fluid Mechanics*, 23(2), 407-428 (2022).
14. **Grace, A.P.**, Stastna, M., Lamb, K.G. and Scott, K.A. Numerical simulations of the three-dimensionalization of a shear flow in radiatively forced cold water below the density maximum. *Physical Review Fluids* 7(2) (2022): 023501 (**Editor's Highlight**)
15. Legare, S., **Grace, A.**, and Stastna, M. Double-Diffusive Instability in a Thin Vertical Channel. *Physics of Fluids*, 33, 114106 (2021)
16. **Grace, A.P.**, Stastna, M., Lamb, K. G., and Scott, K. A. Asymmetries in gravity currents attributed to the nonlinear equation of state. *Journal of Fluid Mechanics*, 915. (2021)
17. **Grace, A.P.**, Stastna, M., and Poulin, F.J., Numerical simulations of the shear instability and subsequent degeneration of basin scale internal standing waves. *Physical Review Fluids* 4.1 (2019): 014802. (**Editor's Highlight**)

Under Review

1. Stastna M., **Grace, A.P.**, On the arrested development of the Rayleigh Taylor instability in the cabbelling regime. *Submitted to Physical Review Fluids*

In Preparation

1. **Grace, A.P.**, and Richter, D.H. On Eulerian anisotropy of settling Lagrangian particles in a turbulent boundary layer.

Awards, Fellowships, & Grants

Postdoctoral Fellowship

NATURAL SCIENCES AND ENGINEERING RESEARCH COUNCIL

January 2025 - August 2025

Partial Postdoctoral Fellowship

NOTRE DAME ENVIRONMENTAL CHANGE INITIATIVE

September 2023-August 2025

Tertia M.C. Hughes Graduate Student Prize

CANADIAN METEOROLOGICAL AND OCEANOGRAPHIC SOCIETY

June 2023

Teaching

Instructor of Record

- DIFFERENTIAL EQUATIONS WITH LINEAR ALGEBRA (CU BOULDER)
- HYDRAULICS LAB (UNIVERSITY OF NOTRE DAME)

January 2025-April 2026

Lecturer

CALCULUS II

January 2021 - April 2021

Teaching Assistant

- INTRODUCTION TO FLUID DYNAMICS
- ASYMPTOTICS AND PERTURBATION THEORY
- CONTINUUM MECHANICS
- CALCULUS
- INTRODUCTION TO COMPUTATIONAL MATHEMATICS
- INTRODUCTION TO THEORETICAL MECHANICS

2016-2022

Mentoring

PH.D ADVISOR: EMMA OPPER

2023-2025

MENTOR: KRISTIN SWARTZ-SCHULT (PHD), NICO CASTRO-FOLKER (PHD), DONOVAN ALLUM (PHD), SIERRA LEGARE (M.MATH), ALEXANDER FOGAL (UG) AND TRAVIS ROBINSON (UG).

2019-Present

APS DIVISION OF FLUID DYNAMICS (DFD) MENTORING PROGRAM

2023-2025

Academic Service

Panelist

ORAL COMMUNICATION IN ACADEMIA AND MATHEMATICS: RESEARCH PRESENTATIONS AND TEACHING

2025

Peer Reviewer

PEER REVIEWER OF OVER 20 PAPERS.

2023-Present

Graduate Committee Member

DEPARTMENT OF APPLIED MATHEMATICS
UNIVERSITY OF COLORADO BOULDER

2025-Present

Session Chair

- APS-DFD: PARTICLE LADEN CONVECTION
- PHYSICAL PROCESSES IN NATURAL WATERS

2022-2024

Award Judging

OUTSTANDING STUDENT PRESENTATION AWARDS JUDGE

2023-2024

Professional Development

Notre Dame Future Faculty Workshop

UNIVERSITY OF NOTRE DAME

May 2023

- Workshop comprised of several panels of current Notre Dame faculty answering questions and providing their experience and expertise in the faculty hiring process.
- Panels included information on writing effective research and teaching statements, how to construct an effective job talk, as well as presentations from faculty members outside of the Notre Dame community.

Compute Canada Deep Learning Seminars

DIGITAL RESEARCH ALLIANCE OF CANADA

June 2021

- Two-day seminar series lead by specialists at the Digital Research Alliance of Canada (formerly Compute Canada) focused on fundamentals and practical applications of machine learning using the PyTorch software package.
- Components included several assignments and labs which included training simple models on linear and nonlinear regression, as well as image classification.

Fluid Dynamics of Sustainability and the Environment

ÉCOLE POLYTECHNIQUE AND UNIVERSITY OF CAMBRIDGE

July 2019

- International summer school consisting of lectures led by world-renowned experts in atmospheric sciences, oceanography, renewable energies, and climate science.
- Summer school featured two poster presentations, and dedicated physical and numerical laboratory work.

Compute Canada High Performance Computing Summer School

COMPUTE CANADA

June 2017

- Two-day summer school consisted of a series of short courses focused on the understanding and efficient use of Digital Research Alliance of Canada (formerly Compute Canada) infrastructure.
- Content included introduction to parallel programming using C++ and FORTRAN, profiling and optimizing code for use on multiple machines, as well as data visualization using Paraview and VisIt.

PROFESSIONAL MEMBERSHIPS

MEMBER: APS DIVISION OF FLUID DYNAMICS

2017-Present

- Branch of the American Physical Society dedicated to advancement of fluid physics.

MEMBER: AMERICAN GEOPHYSICAL UNION

2023-Present

- Organization dedicated to advancing Earth and Space science.

MEMBER: THE WATER INSTITUTE

2018-2022

- Canadian leader in interdisciplinary water related research.

MEMBER: GLOBAL WATER FUTURES

2018-2022

- Multi-institute organization with the aim of positioning Canada as a world leader in water and cold regions research.

Special Seminars

Modeling the influence of buoyancy: implications for mixing and transport in natural waters and the atmosphere.

Boulder, Colorado

INVITED TALK: BOULDER FLUIDS AND THERMAL SCIENCES SEMINAR

November, 2025

Eulerian and Lagrangian transport in wall-bounded turbulent flows

Boulder, Colorado

INVITED TALK: UNIVERSITY OF COLORADO BOULDER, DEPARTMENT OF APPLIED MATHEMATICS
COLLOQUIUM

August, 2025

Environmental Fluid Flows: Stratification, Particles, and Turbulence

Boulder, Colorado

INVITED TALK: UNIVERSITY OF COLORADO BOULDER, DEPARTMENT OF APPLIED MATHEMATICS

February, 2025

Mathematical modelling to gain insights into the environment around us

Charlottetown, P.E.I.

INVITED TALK: UNIVERSITY OF PRINCE EDWARD ISLAND, DEPARTMENT OF MATHEMATICS AND
COMPUTATIONAL SCIENCES

October, 2024

Simulations and scaling of density driven flows in the cabbelling regime

Peterborough, Ontario

INVITED TALK: TRENT UNIVERSITY PHYSICS AND ASTRONOMY SEMINAR SERIES

September 2022

A Tour of the Incompressible Navier-Stokes Equations

GRADUATE STUDENT DEPARTMENT SEMINAR: UNIVERSITY OF WATERLOO DEPARTMENT OF
APPLIED MATHEMATICS

Online

September 2020

A brief introduction to geophysical fluid dynamics

INVITED TALK: TRENT MATHEMATICS AND STATISTICS CONFERENCE

Peterborough, Ontario

March 2019

Conference Presentations

Exploring Removal Mechanisms in the Pi Convection-Cloud Chamber

SWARTZ-SCHULT ET. AL.

Contributed Talk: 106th American Meteorological Society Annual Meeting

Houston, Texas

January 2026

Multi-scale interactions in turbulent mixed convection drive efficient transport of Lagrangian particles.

GRACE, A.P., AND RICHTER, D.H.

Contributed Talk: American Physical Society Division of Fluid Dynamics

Houston, Texas

November 2025

Multi-scale interactions in turbulent mixed convection drive efficient transport of Lagrangian particles.

GRACE, A.P., AND RICHTER, D.H.

Interact Session: American Physical Society Division of Fluid Dynamics

Salt Lake City, Utah

November 2024

On the arrested development of the Rayleigh Taylor instability with and without cabbeling.

STASTNA, M., AND GRACE, A.P.

Contributed Talk: American Physical Society Division of Fluid Dynamics

Salt Lake City, Utah

November 2024

The Settling Rates of Particles in Rayleigh-Bénard Turbulence.

SWARZ-SCHULT, K., ANDERSON, J.C., SADI, H.F., SULAIMAN, S.V., CANTRELL, W., SHAW R.A.,
RICHTER, D.H., AND GRACE, A.P.

Contributed Talk: American Physical Society Division of Fluid Dynamics

Salt Lake City, Utah

November 2024

Slip velocity statistics of settling inertial particles in wall bounded turbulence.

GRACE, A.P., BERK, T., BRAGG, A.D., AND RICHTER, D.H.

Contributed Talk: American Physical Society Division of Fluid Dynamics

Salt Lake City, Utah

November 2024

Restratification in late winter lakes induced by cabbeling

GRACE, A.P., FOGAL, A., AND STASTNA, M.

Contributed Talk: American Geophysical Union Annual Meeting

San Francisco, California

December 2023

Phase space modeling for settling enhancement of inertial particles

GRACE, A.P., RICHTER, D.H., AND BRAGG, A.

Contributed Talk: American Geophysical Union Annual Meeting

San Francisco, California

December 2023

Phase space modeling for settling enhancement of inertial particles

GRACE, A.P., RICHTER, D.H., AND BRAGG, A.

Contributed Talk: American Physical Society Division of Fluid Dynamics

Washington, District of Columbia

November 2023

PDF-based model for inertial particles settling in a turbulent boundary layer using an asymptotic closure approximation

ZHANG, Y., GRACE, A.P., RICHTER, D.H., AND BRAGG, A.

Contributed Talk: American Physical Society Division of Fluid Dynamics

Washington, District of Columbia

November 2023

- Three-dimensional simulations of gravity currents in cold, fresh water**
 CASTRO-FOLKER, N., **GRACE, A.P.**, AND STASTNA, M.
 Contributed Talk: American Physical Society Division of Fluid Dynamics
 Washington, District of Columbia
 November 2023
- Gravity Currents in the Cabbeling Regime**
GRACE, A.P., STASTNA, M., LAMB, K.G., AND SCOTT, K.A.
 Contributed Talk: Canadian Mathematical Society Winter Meeting
 Toronto, Ontario
 December 2022
- Gravity Currents in the Cabbeling Regime**
GRACE, A.P., STASTNA, M., LAMB, K.G., AND SCOTT, K.A.
 Contributed Talk: American Physical Society Division of Fluid Dynamics
 Indianapolis, Indiana
 November 2022
- 3D Simulations of the Interior of an Ice-Covered Lake subjected to Spatially Heterogeneous Solar Radiation Intensity**
 ALLUM, D.J.M., **GRACE, A.P.**, AND STASTNA, M.
 Contributed Talk: American Physical Society Division of Fluid Dynamics
 Indianapolis, Indiana
 November 2022
- Double Diffusive Instability with a Constriction**
 LEGARE, S., **GRACE, A.P.**, AND STASTNA, M.
 Contributed Talk: American Physical Society Division of Fluid Dynamics
 Indianapolis, Indiana
 November 2022
- Gravity Currents in the Cabbeling Regime**
GRACE, A.P., STASTNA, M., LAMB, K.G., AND SCOTT, K.A.
 Contributed Talk: Physical Processes in Natural Waters
 Vancouver, British Columbia
 July 2022
- Shear mediates downward heat fluxes in unstably stratified environments.**
GRACE, A.P., STASTNA M., LAMB, K.G., AND SCOTT, K.A.
 Contributed Talk: The Canadian Meteorological and Oceanographic Society Congress
 Online
 June 2021
- Gravity current evolution in the cold water regime**
GRACE, A.P., STASTNA M., LAMB, K.G., AND SCOTT, K.A.
 Contributed Talk: The Canadian Meteorological and Oceanographic Society Congress.
 Online
 June 2020
- Numerical modeling of hydrodynamic-ice dynamic coupling in a small lake**
GRACE, A.P., STASTNA M., LAMB, K.G., AND SCOTT, K.A.
 Contributed Talk: International Association for Great Lakes Research
 Brockport, New York
 May 2019
- Simulations of motion in the bottom boundary layer induced by an internal standing wave**
GRACE, A.P., STASTNA M.
 Contributed Talk: American Physical Society Division of Fluid Dynamics
 Atlanta, Georgia
 November 2018
- Reconciling simulations of theoretical and experimental laboratory scale seiche motion**
GRACE, A.P., STASTNA, M., AND POULIN, F.J.
 Contributed Talk: European Geophysics Union General Assembly, Vienna, Austria.
 Vienna, Austria
 April 2018
- The response of a basin scale seiche due to variation of the aspect ratio of the density surface**
GRACE, A.P., STASTNA, M., AND POULIN, F.J.
 Contributed Talk: American Physical Society Division of Fluid Dynamics
 Denver, Colorado
 November 2017

Poster Presentations _____

Multi-scale interactions in turbulent mixed convection drive efficient transport of Lagrangian particles.

GRACE, A.P., AND RICHTER, D.H.

Contributed Poster: American Geophysical Union Annual Meeting

New Orleans, Louisiana

December 2025

Using Upscaling Techniques to Incorporate Particle Settling Speed Enhancement in a Coarse-Mode Deposition Model.

GRACE, A.P., AND RICHTER, D.H.

Contributed Poster: American Geophysical Union Annual Meeting

New Orleans, Louisiana

December 2025

Volumetric heating models and their impact on small scale, non-hydrostatic dynamics

GRACE, A.P., STASTNA, M., LAMB, K.G., AND SCOTT, K.A.

Contributed Poster: Global Water Futures Annual Science Meeting

Online

May 2020

Numerical process studies of the circulation and ice formation on a small lake.

GRACE, A.P., STASTNA, M., LAMB, K.G., AND SCOTT, K.A.

Contributed Poster: The Fluid Dynamics of Sustainability and the Environment, École Polytechnique

Palaiseau, France.

July 2019

Modeling ice growth in Marginal Ice Zones using a coupled hydrodynamic-ice model

GRACE, A.P., STASTNA, M., LAMB, K.G., AND SCOTT, K.A.

Contributed Poster: Global Water Futures Annual Science Meeting

Saskatoon, Saskatchewan

May 2019

From DNS of shear instability to cross-layer transport

GRACE, A.P., STASTNA, M., AND POULIN, F.J.

Contributed Poster: The Canadian Meteorological and Oceanographic Society Congress.

Toronto, Ontario

June 2017